

A new look at gray-matter decreases in chronic pain

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Word count: 1367 words (excluding references), 12 references

Scientific views on chronic pain have evolved dramatically over the past decades. It is increasingly clear that many types of chronic pain are promoted and maintained by changes in the brain that can occur after injury or insult to the body. These include remodeling of neural pathways, glial changes, and neuroinflammation (Tan and Kuner 2021). Central nervous system (CNS) alterations can cause pain to spread beyond an initial site of injury, making it a systemic rather than a localized disorder. Co-occurring pain across body sites and etiologies is linked to shared genetic susceptibilities and common functional changes in the brain (Zorina-Lichtenwalter et al. 2023; Tanguay-Sabourin et al. 2023). Given that chronic pain is a strong risk factor for other mental health conditions (Tanguay-Sabourin et al. 2023), including depression, anxiety, substance use disorders, and suicide, it is critical to understand structural and functional brain changes with chronic pain and their relationships with these co-occurring conditions. A recent paper by Bhatt et al. (Bhatt et al. 2023) takes an important step forward in addressing this need in one of the largest, well-powered, and most convincing analyses of brain gray-matter changes in chronic pain to date.

One crucial piece of the puzzle of chronic pain lies in brain structural changes in human MRI studies. Early studies reported dramatic reductions in MRI measures of gray-matter density linked to chronic pain (Apkarian et al. 2004), which have been interpreted in terms of damage to emotional and self-regulatory systems. Such brain alterations could help to explain chronic pain. For example, they could create a positive feedback loop between pain and psychological/behavioral changes that support it, including fear, depression, and withdrawal. However, these effects were identified in small, underpowered studies, with inevitable confounds such as substance and medication use, body fat, mood disorders, and head movement. In addition, while early meta-analyses identified consistent gray-matter changes, more recent, larger ones have failed to find evidence for consistent changes at the statistical thresholds needed to correct for multiple comparisons across the brain (Henn et al. 2023). Nevertheless, they suggest there may be gray-matter decreases in the medial prefrontal cortex and anterior insula in particular (Henn et al. 2023) – regions important for self-regulation and influences on behavior and bodily physiology based on cognitive appraisals. Thus, it remains to be determined (1) whether brain gray-matter is altered with chronic pain; (2) if so, which regions are affected in which pain conditions; (3) whether these effects are driven by other co-occurring variables; and (4) whether these effects mediate the relationship between chronic pain and co-occurring mental health and substance use disorders.

To shed light on these questions, the recent paper by Bhatt and colleagues (Bhatt et al. 2023) analyzed data from the UK Biobank to compare MRI measures of regional gray matter in 10,984 participants with chronic pain and 10,984 pain-free controls that were carefully matched on age and sex. They found widespread reductions in cortical surface area (but not thickness or subcortical volume) across conditions on average and an increase in cortical thickness in the precuneus and lateral orbitofrontal cortices. Localized chronic pain was associated most strongly with reductions in somatosensory, motor, dorsal anterior insula, and posterior medial (i.e., posterior cingulate) regions. These include many of the regions most strongly associated with sensory and interoceptive processing. Multisite chronic pain was associated with stronger reductions overall and particularly with reductions in medial prefrontal, lateral fronto-parietal, and lateral temporal cortices. Though these patterns were not directly statistically compared, the different patterns are suggestive of changes in somatosensory systems with both localized and widespread pain and additional changes in systems associated with higher-level context-based regulation of cognition and behavior for widespread pain. Somatosensory, motor, and medial prefrontal cortices are the origins of descending pathways that can modify how nociceptive (pain-related) signals are processed in the brainstem and spinal cord (Tan and Kuner 2021), suggesting that chronic pain may involve disruption of closed-loop feedback circuits that normally inhibit pain, creating a pain-facilitatory shift (Ossipov, Morimura, and Porreca 2014).

In addition to more careful matching of controls than previous studies, the sample size (4 times as large as the total sample size contributing to the largest previous meta-analysis (Henn et al. 2023) allowed Bhatt et al. to analyze changes associated with pain both unique to specific body sites and common across body sites, as well as associations with suicide. Regional surface area reductions were most strongly associated with chronic musculoskeletal pain, including back, neck, knee, and hip pain. No reductions with headaches or abdominal pain were reported. Instead, headaches stood out as associated with widespread *increases* in cortical thickness (and no decreases), and abdominal pain stood out by being the only condition to be associated with subcortical volume differences in left amygdala, brainstem, and cerebellum. These patterns were not directly compared, and the effect sizes may be too small to allow statistically robust comparisons even in this large sample. Nevertheless, the results suggest that a ‘one size fits all’ approach is insufficient to describe pain-associated gray matter changes, and that different supra-nociceptive mechanisms may contribute to maintenance of different forms of chronic pain.

The empirical foundation provided by Bhatt et al.'s study paves the way for future studies to address important unanswered questions. What is the functional significance of these changes? Do they constitute “damage” to the brain, as some have claimed, or are they normal adaptations to pain that promote behavioral withdrawal and recovery (Seymour, Crook, and Chen 2023)? Are they related to mental health symptoms, neuroinflammation, and/or the neural remodeling seen after nerve injury in nonhuman animals? Are they related to accelerated aging, and do they interact with other factors that promote premature aging (Cruz-Almeida et al. 2019)?

One unique contribution of Bhatt et al.'s study is in understanding relationships with suicide, which is strongly related to pain and depression. Among the areas with pain-linked gray matter changes, precuneus emerged as a mediator of suicide attempts. Along with the ventromedial prefrontal cortex, the precuneus may be part of a system that is responsible self-referential thought and construction of feelings, and which undergoes a shift towards negative internal states in those at risk for suicide (Schmaal et al. 2020). Studies of direct brain intervention in the precuneus and inter-connected regions would provide a way of both validating this association and translating neuroimaging findings to clinical interventions.

Even with the relatively careful controls in Bhatt et al.'s study, there are important issues that must be addressed in future studies. As Bhatt et al. note, their findings do not address whether structural changes play a causal role in contributing to pain and related mental health and substance use problems. While a few small studies have suggested gray matter changes are reversible with successful treatment (e.g., (Seminowicz et al. 2011)), the magnitude of treatment effects is unknown. Even if changes are reversible, they may still play a causal role in maintaining pain and dysfunction. In addition, the effects Bhatt et al. identify are small, and it is unclear whether there are reproducible differences across body sites and pain conditions. It is possible that the different patterns of regional associations they report are artifacts of thresholding small effects and are not reliably different. Furthermore, though the study's sample matching process and exclusion of people with mental health conditions and obesity represents a valiant effort to control for some confounders, it is possible that some of the observed changes are related to differences in brain water content (Pomares et al. 2017), head movement (which can blur local signals), and other factors related to BMI, stress, mental health, and use of medications and other drugs. Future analyses assessing whether these other variables are plausible mediators, and converging evidence from animal models, pain treatment studies,

direct neurological interventions, and quantitative MRI (Pomares et al. 2017) will help clarify the role of pain *per se* in altering brain structure.

In summary, Bhatt et al. have provided some of the most comprehensive, large-sample evidence for gray-matter associations with pain to date, with potential differences across body sites and associations with suicide risk. The study represents an important step in understanding how the brain changes with, and contributes to, chronic pain, and provides a launching point for future studies using converging methodologies.

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